

ISHAN DURUGKAR

Senior Research Scientist, Sony AI • Reinforcement Learning

@ ishand@cs.utexas.edu

idurugkar.github.io

Google Scholar

github.com/idurugkar

in in/idurugkar

SUMMARY

I am a research scientist on Sony AI's Game AI team, where I train and deploy reinforcement learning agents in AAA video games under real-time, human-in-the-loop constraints. I also publish research on RL algorithms at NeurIPS, ICML, ICLR, and AAAI. I completed my Ph.D. at the University of Texas at Austin, advised by Prof. Peter Stone; my thesis studied how RL algorithms can be improved by estimating and controlling the data distributions they induce. My research interests span deep, multi-agent, imitation, goal-conditioned, and unsupervised RL. Recently, I have been investigating how RL combines with foundation models: LLM and agentic interfaces that make learned policies more accessible, world models built on vision-language models, and LLM post-training methods studied through the distributions they induce.

SELECTED HIGHLIGHTS

- **RL in production-scale games.** I train RL agents to accomplish complex tasks in (undisclosed) AAA Sony titles, on the same Sony AI Game AI team that built *GT Sophy*, the champion-level Gran Turismo agent.
- **Foundation models.** I built MCP tools and an LLM interface that let users direct trained RL policies in-game, and fine-tuned a vision-language-action (VLA) model for gameplay. I am a co-author of *Semantic World Models* (2025, with collaborators at the University of Washington), which fine-tunes a VLM to answer questions about future states instead of reconstructing pixels.
- **Research record.** *Generative Multi-Adversarial Networks* (ICLR 2017) is an early and widely cited multi-adversary GAN framework. First-author papers at NeurIPS, ICLR, and AAAI.

RESEARCH & WORK EXPERIENCE

Research Scientist

SONY AI — GAME AI

2023 – present

The Game AI team conducts fundamental RL research and applies it to create more capable AI agents for video games. I extract features from games, design rewards, and tune algorithms to train RL agents that accomplish complex tasks in (undisclosed) AAA titles, and I contribute to the team's research on what learning agents can do in these massive games. I am also responsible for the path from research to demo: I optimize policies to run in real time alongside a live game, make them robust to noisy interaction with people, and evaluate and select the policy snapshots that are presented. Most recently, I built MCP tools and an LLM interface to direct trained RL policies in-game, and I evaluate new techniques for game-playing agents, including VLA models, world models, and LLM harnesses for agentic workflows.

Research Scientist Intern

DEEPMIND

Summer 2021

Worked with Volodymyr Mnih on the Discovery team. Studied unsupervised RL: learning skillful policies in the absence of reward signals from the environment, with the learned skills evaluated on the Atari benchmark.

Research Intern

MICROSOFT RESEARCH (MSR)

Summer 2018

Worked with Adith Swaminathan and Matthew Hausknecht on making RL agents easier to develop by incorporating multiple feedback channels into policy learning. In particular, used demonstrations to speed up the learning of a policy-gradient agent.

Graduate Research Assistant

THE UNIVERSITY OF TEXAS AT AUSTIN

2017–2023

Ph.D. research advised by Peter Stone (LARG). Primary focus on Reinforcement Learning, with projects on optimization of TD learning, human feedback, sim-to-real transfer, multi-agent RL, and goal-conditioned RL. Doctoral thesis: improving RL algorithms by estimating and controlling the visitation distributions induced by the agent.

Research Assistant

UNIVERSITY OF MASSACHUSETTS AT AMHERST

2015–2017

Research with Sridhar Mahadevan (ALL) on: (1) Generative Adversarial Networks (GANs) and Variational Auto-Encoders (VAEs), (2) temporal abstractions for actions in Deep RL, (3) manifold learning on graphs, and (4) learning policies to navigate large knowledge bases.

Research Member

MAHARASHTRA INSTITUTE OF TECHNOLOGY

2013

Research on non-convex optimization using swarm techniques.

PERSONAL PROJECTS

LLM post-training through visitation distributions

PERSONAL PROJECT (PRIVATE REPOSITORY)

Ongoing

I am investigating LLM post-training methods (supervised fine-tuning, RLHF, DPO, GRPO, OPD) through the lens of distribution matching on open-weights model like Llama-3.1 and Qwen3. The aim is to identify effective post-training techniques through the visitation-distribution analysis from my thesis to these methods.

SELECTED PUBLICATIONS

(**bold** = author; see the **Publications** section below for the full list)

- Jacob Berg, Chuning Zhu, Yanda Bao, **Ishan Durugkar**, Abhishek Gupta. “Semantic World Models.” *arXiv:2510.19818*, 2025.
- Caroline Wang, Muhammad A. Rahman, **Ishan Durugkar**, Elad Liebman, Peter Stone. “N-Agent Ad Hoc Teamwork.” **NeurIPS** 2024.
- Siddhant Agarwal, **Ishan Durugkar**, Peter Stone, Amy Zhang. “f-Policy Gradients: A General Framework for Goal-Conditioned RL using f-Divergences.” **NeurIPS** 2023.
- Caroline Wang, **Ishan Durugkar**, Elad Liebman, Peter Stone. “DM²: Distributed Multi-Agent Reinforcement Learning via Distribution Matching.” **AAAI** 2023.
- **Ishan Durugkar**, Mauricio Tec, Scott Niekum, Peter Stone. “Adversarial Intrinsic Motivation for Reinforcement Learning.” **NeurIPS** 2021.
- Siddharth Desai, **Ishan Durugkar**, Haresh Karnan, Garrett Warnell, Josiah Hanna, Peter Stone. “An Imitation from Observation Approach to Transfer Learning with Dynamics Mismatch.” **NeurIPS** 2020. (*joint first author*)
- Brahma Pavse, **Ishan Durugkar**, Josiah Hanna, Peter Stone. “Reducing Sampling Error in Batch Temporal Difference Learning.” **ICML** 2020.
- Rajarshi Das, Shehzaad Dhuliawala, Manzil Zaheer, Luke Vilnis, **Ishan Durugkar**, Akshay Krishnamurthy, Alex Smola, Andrew McCallum. “Go for a Walk and Arrive at the Answer: Reasoning over Paths in Knowledge Bases using RL.” **ICLR** 2018.
- **Ishan Durugkar**, Ian Gemp, Sridhar Mahadevan. “Generative Multi-Adversarial Networks.” **ICLR** 2017.

EDUCATION

PH.D. COMPUTER SCIENCE (2017 – 2023)

THE UNIVERSITY OF TEXAS AT AUSTIN, USA

Research areas: Reinforcement Learning, Multi-agent Reinforcement Learning, Robotics, Imitation Learning, Deep Learning

Affiliations: Peter Stone, Learning Agents Research Group (LARG)

M.S. COMPUTER SCIENCE (2014 – 2017)

UNIVERSITY OF MASSACHUSETTS AMHERST

Research Areas: Deep Learning, Reinforcement Learning, Generative Adversarial Networks

GPA: 3.97

Affiliations: Sridhar Mahadevan, Autonomous Learning Lab (ALL)

B.ENG. COMPUTER ENGINEERING (2007 – 2013)

MAHARASHTRA INSTITUTE OF TECHNOLOGY, INDIA

Research and Extracurricular interests: Robotics, Swarm techniques, non-convex optimization

SKILLS

Reinforcement learning: deep RL, multi-agent RL, imitation & inverse RL, goal-conditioned RL, unsupervised RL / skill discovery, sim-to-real transfer; reward design and RL-policy deployment.

LLMs & agents: LLM-driven control of RL policies, MCP tool building, agentic interfaces; VLA fine-tuning; LLM post-training (SFT, RLHF, DPO, GRPO).

Engineering: Python (proficient); C++ (intermediate); PyTorch, JAX, TensorFlow (and Theano); large-scale / distributed NN training (Slurm, HTCondor, and proprietary systems); HPC clusters.

RESEARCH STATEMENT

I am a Research Scientist at Sony AI working in the Game AI team. We focus on fundamental **Reinforcement Learning** (RL) research and apply techniques to create more capable AI agents for video games. Challenges we focus on include RL training techniques for agents that can be deployed on consumer PlayStation devices, are predictable and enjoyable to interact with, and empower developers to create the next generation of video games.

I also contribute to research threads focusing on **multi-agent reinforcement learning**, **robotics**, and **goal-conditioned reinforcement learning**, with collaborators at UT Austin and the University of Washington. Most recently, with collaborators at the University of Washington, I have worked on **world models built on vision-language models**: *Semantic World Models* (2025) fine-tunes a VLM to answer questions about future states instead of reconstructing pixels, and uses it for planning in robotics tasks. In parallel, as a personal project, I am studying **LLM post-training** methods through the same distributional lens. I completed my Ph.D. from the

Department of Computer Science at the University of Texas at Austin, advised by Prof. Peter Stone. My doctoral thesis focused on improving RL algorithms by controlling the data they generate — in particular, the estimation and control of the visitation distributions induced by RL agents, applied to imitation learning, goal-conditioned RL, sim-to-real transfer, and multi-agent RL.

My previous industry experience has been in research internships at **DeepMind** in 2021 and **Microsoft Research** in 2018.

OTHER WORK EXPERIENCE

Software Engineering Intern

AMAZON COMIXOLOGY, NEW YORK CITY

Summer 2015

Implemented components of the Comixology continuous integration pipeline. Developed a build system based on Ruby, PHP and ANT that enabled developers to run Unit, Integration and Functional Tests in a single setting, and integrated it into the development process.

Data Scientist – Research Assistant

UMASS AMHERST ALUMNI ASSOCIATION

2015 – 2016

Used data aggregation, cleaning and analysis along with machine learning techniques to identify and better engage alumni with on-campus events and fund raising.

Software Engineer

MICROSOFT INDIA, HYDERABAD

2013 – 2014

As part of the Code Search team in Visual Studio Online, implemented a prototype of code search based on Elasticsearch, which was then approved for further development. Participated in the effort to scale the prototype to integrate with Visual Studio Online.

ROBOTICS COMPETITIONS

RoboCup Standard Platform League

UT AUSTIN VILLA

2019 - 2022

Quarter-finalists 2019, 2022, 4th place in 2021

ABU Asia-Pacific Robot Contest

MAHARASHTRA INSTITUTE OF TECHNOLOGY

2010 – 2012

Represented India in 2010 and 2012

TEACHING

Teaching Assistant

THE UNIVERSITY OF TEXAS AT AUSTIN

Fall 2019

Reinforcement Learning: Theory and Practice

Teaching Assistant

UMASS AMHERST

2015 – 2017

- Machine Learning

- Distributed and Operating Systems

SERVICE

Area Chair

ICML, RLC

2025

Organizer – GCRL Workshop

NEURIPS

Fall 2024

Member of the organizing committee for the goal-conditioned RL workshop at NeurIPS 2023 <https://goal-conditioned-rl.github.io/2023/>

RoboCup Standard Platform League Organizing Committee

UT AUSTIN

Fall 2022 - Spring 2023

Member of the organizing committee for the standard platform league (SPL) at RoboCup 2023 to be held in Bordeaux, France <https://spl.robocup.org/committees-2023/>

International Joint Conference on Artificial Intelligence (IJCAI)

ASSISTANT WORKFLOW CHAIR

2023

RL Reading Group – Organizer

UT AUSTIN

Spring 2019 – Fall 2022

Graduate Admission Committee

UT AUSTIN

Spring 2019

Volunteer

Indian Student Association

UNIVERSITY OF MASSACHUSETTS, AMHERST

Fall 2015 – Fall 2016

Vice-President of the Indian Student Association at the University of Massachusetts, Amherst.

REVIEWING EXPERIENCE

Conferences: AAAI (2021–2023), ICLR (2021–2025, *Highlighted Reviewer* 2022), NeurIPS (2016, 2020–2024), ICML (2021, 2022, 2024, 2025), CoLLAs (2022, *inaugural PC*), AISTATS (2022, 2023), CoRL (2020), IJCAI (2020, 2021), ICRA (2023), RLC (2024 *inaugural PC*, 2025).

Journals: JAIR (2022), JMLR (2023, 2024), TMLR (2022–2024), IEEE TNNLS (2020–2023), Springer Machine Learning (2022–2024).

Workshops: Lifelong Machine Learning (ICML 2020), Multi-Task and Lifelong RL (ICML 2019), Neverending Learning (ICLR 2021).

PUBLICATIONS

Conference Proceedings

- [Wan+24] Caroline Wang et al. “N-agent ad hoc teamwork”. In: *Advances in Neural Information Processing Systems*. Vol. 37. 2024, pp. 111832–111862.
- [Aga+23] Siddhant Agarwal et al. “f-Policy Gradients: A General Framework for Goal-Conditioned RL using f-Divergences”. In: *Advances in Neural Information Processing Systems*. Vol. 36. 2023.
- [Wan+23] Caroline Wang et al. “DM²: Distributed Multi-Agent Reinforcement Learning via Distribution Matching”. In: *Proceedings of the 37th AAAI Conference on Artificial Intelligence*. 2023.
- [Nar+22] Sai Kiran Narayanaswami et al. “Towards a real-time, low-resource, end-to-end object detection pipeline for robot soccer”. In: *Robot World Cup*. **nominated for best paper**. Springer, 2022, pp. 62–74.
- [Dur+21a] Ishan Durugkar et al. “Adversarial Intrinsic Motivation for Reinforcement Learning”. In: *Advances in Neural Information Processing Systems*. 2021.
- [Des+20] Siddharth Desai et al. “An Imitation from Observation Approach to Transfer Learning with Dynamics Mismatch”. In: *Advances in Neural Information Processing Systems*. *Joint first author*. 2020.
- [DLS20] Ishan Durugkar, Elad Liebman, and Peter Stone. “Balancing Individual Preferences and Shared Objectives in Multiagent Reinforcement Learning”. In: *Proceedings of the 29th International Joint Conference on Artificial Intelligence*. 2020, pp. 2505–2511.
- [Pav+20b] Brahma Pavse et al. “Reducing Sampling Error in Batch Temporal Difference Learning”. In: *Proceedings of 37th International Conference on Machine Learning, Vienna, Austria*. 2020.
- [Das+18] Rajarshi Das et al. “Go for a Walk and Arrive at the Answer: Reasoning Over Paths in Knowledge Bases using Reinforcement Learning”. In: *International Conference on Learning Representations*. 2018.
- [DGM17] Ishan Durugkar, Ian Gemp, and Sridhar Mahadevan. “Generative multi-adversarial networks”. In: *International Conference on Learning Representations*. 2017.
- [DS17a] Ishan Durugkar and Peter Stone. “TD Learning with Constrained Gradients”. In: *Proceedings of the Deep Reinforcement Learning Symposium, NIPS 2017*. Long Beach, CA, USA, Dec. 2017.
- [PGD17] Mario Parente, Ian Gemp, and Ishan Durugkar. “Unmixing in the presence of nuisances with deep generative models”. In: *2017 IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*. IEEE. 2017, pp. 5189–5192.
- [Tho+17] Philip S Thomas et al. “Predictive off-policy policy evaluation for nonstationary decision problems, with applications to digital marketing”. In: *Proceedings of the Thirty-First AAAI Conference on Artificial Intelligence*. 2017, pp. 4740–4745.
- [KDK13] Anand J Kulkarni, Ishan P Durugkar, and Mrinal Kumar. “Cohort intelligence: a self supervised learning behavior”. In: *2013 IEEE international conference on systems, man, and cybernetics*. IEEE. 2013, pp. 1396–1400.

Workshops and Symposium

- [Hud+22] Eddy Hudson et al. “ABC: Adversarial Behavioral Cloning for Offline Mode-Seeking Imitation Learning”. arXiv preprint arXiv:2211.04005, Deep RL Workshop and Offline RL workshop at NeurIPS. 2022.
- [Dur+21b] Ishan Durugkar et al. “Wasserstein Distance Maximizing Intrinsic Control”. NeurIPS Deep Reinforcement Learning Workshop. 2021.
- [Pav+20a] Brahma S. Pavse et al. “On Sampling Error in Batch Action-Value Prediction Algorithms”. In the Offline Reinforcement Learning Workshop at Neural Information Processing Systems (NeurIPS), December 2020. Dec. 2020.
- [Dur+19] Ishan Durugkar et al. “Multi-Preference Actor Critic”. The Multi-disciplinary Conference on Reinforcement Learning and Decision Making. 2019.
- [DS18] Ishan Durugkar and Peter Stone. “Adversarial goal generation for intrinsic motivation”. AAAI Student Abstracts. 2018.
- [DS17b] Ishan Durugkar and Peter Stone. “TD learning with constrained gradients”. NeurIPS Deep Reinforcement Learning Symposium. 2017.

Preprints

- [Ber+25] Jacob Berg et al. *Semantic World Models*. 2025.

- [Wan+25] Caroline Wang et al. *Sequence Modeling for N-Agent Ad Hoc Teamwork*. 2025.
- [Dur+16] Ishan P Durugkar et al. *Deep reinforcement learning with macro-actions*. arXiv preprint 1606.04615. 2016.
- [Gem+16a] Ian Gemp et al. *Deep Generative Models for Spectroscopic Analysis on Mars*. ResearchGate. 2016.
- [Gem+16b] Ian Gemp et al. *Inverting variational autoencoders for improved generative accuracy*. arXiv preprint 1608.05983. 2016.